

Two Cacao Varieties of Pará, Brazil

Field-video verification + research dossier — grafted CCN-51 “Ponta Verde” vs common (traditional) cacao

Source footage: 4 clips from a cacao farm in Pará (IMG_7654 / IMG_7669 / IMG_7672 / IMG_7673, ~101 s). The farmers identify two varieties: a **grafted CCN-51 “Ponta Verde”** that fruits year-round, and **“essas outras cabaças” / common cacao** with a seasonal cycle. Every photo below is colour-labelled so you can see at a glance which variety it shows.

1 · Colour key — how to read the photos

VARIETY A vs VARIETY B SIDE BY SIDE ON THE SAME FARM	CCN-51 “Ponta Verde” (grafted) — <i>smooth</i> pods, shallow ribbing, blunt tip, yellow when ripe, fruits all year
VARIETY B COMMON CACAO (TRADITIONAL)	Common / traditional cacao — <i>deeply furrowed</i> (8–10 ridges), pointed tip, red-purple when ripe, seasonal

2 · Variety A — CCN-51 “Ponta Verde” (grafted, year-round)

VARIETY A vs VARIETY B SIDE BY SIDE ON THE SAME FARM



IMG_7669 — the tree the farmer says fruits “every month of the year”. All pods smooth, shallow-ribbed, blunt-tipped, green → yellow. This is the A-type phenotype.

VARIETY A vs VARIETY B SIDE BY SIDE ON THE SAME FARM



IMG_7672 — the large SMOOTH, bright-YELLOW pod with the black basal patch. This is the pod the farmer used to explain pest vs disease. A-type: smooth + yellow when ripe.

VARIETY A vs VARIETY B SIDE BY SIDE ON THE SAME FARM

A CCN-51 'PONTA VERDE' — POD ON TREE



IMG_7672 — very large smooth yellow-green pod on the tree; matches the A-type pods seen throughout the clips.

3 · Variety B — common / traditional cacao (seasonal)

VARIETY B
COMMON CACAO (TRADITIONAL)

B COMMON CACAO — FURROWED RED-PURPLE POD



IMG_7654 — close-up of the B-type pod: DEEPLY FURROWED (8–10 ridges), crimson-purple, pointed tip. The farmer calls these “essas outras cabaças” — the seasonal common cacao.

4 · Both varieties in one frame — the direct contrast

VARIETY A vs VARIETY B
SIDE BY SIDE ON THE SAME FARM



IMG_7654 — the farmer holds a furrowed red-purple pod (B) against a branch of smooth green pods (A): the two phenotypes on the same farm, seconds apart.

5 · Verified field comparison

Trait	Variety A — CCN-51 “Ponta Verde” (grafted)	Variety B — common / traditional cacao
Pod surface	Smooth, very shallow ribbing	Deeply furrowed, 8–10 pronounced ridges
Pod tip	Blunt / rounded	Distinctly pointed
Ripe colour	Bright yellow (occasional black basal patch)	Orange-red → crimson-purple
Pod size	Large (~15–18 cm)	Slightly smaller, more elongate
Fruiting rhythm	Year-round — pods at every stage at any visit	Strongly seasonal — synchronous waves, fruitless period
Propagation	Grafted clone (enxertia de duas plantas)	Seed-planted (traditional / forastero-type)
Farmer's words	“sempre vai ter frutos... durante todo o ano”	“um período que não tem fruto nenhum”

6 · Variety A — CCN-51 “Ponta Verde”: history & agronomy

Origins (Ecuador, 1960s–1980s): CCN-51 (Colección Castro Naranjal 51) was bred by Ecuadorian agronomist Homero Castro over ~12 years of crossing and selection. The “51” marks the 51st cross in the series; the clone derives from a 3-way cross involving IMC 67 / ICS 95 / O-1 in its pedigree. First production reports date to 1965 (Finca Sofia, Naranjal); the variety was released publicly in 1984. It was bred as a crisis response to Ecuador's witches' broom (*Moniliophthora perniciosa*) and Ceratocystis wilt (mal del machete), and for high, reliable yield. Castro died in the late 1980s before patenting it, so the clone spread freely.

Genetics: genetic studies put CCN-51 at roughly 1.1% Nacional, 45.4% IMC, 22.2% Criollo, 21.5% Amelonado, 3.9% Contamana, 2.5% Purús and 2.1% Marañon — essentially a *forastero*-dominant hybrid with a thin Nacional fraction. Not fine-flavour or heirloom; the industry classifies it as “bulk cacao”. ~70% of Ecuadorian production and ~90% of new plantings there over the last two decades; now spreading across Latin America and Africa.

Agronomy: yields ~700–1,100 kg/ha on farms (vs 300–400 kg/ha for Nacional) and up to 2–3 t/ha under ideal conditions; Brazilian field trials recorded 2,143 kg/ha (year 6) and 2,955 kg/ha (year 7). Self-compatible, grows in full sun, tolerates a wider disease load, starts producing ~2 years after planting — the traits behind year-round fruiting. Registered among clonal cultivars (RNC/MAPA) in Brazil; common grafting choice in the Amazon (Pará) and Bahia.

“Ponta Verde”: the name used by the farmers — not an official registry denomination. Most plausibly a local/biotype naming of CCN-51 in Pará (green-tipped pods — “ponta verde”), i.e. the grafted CCN-51 clone as known on the farm.

7 · Variety B — common / traditional Pará cacao: history & identity

Amazonian origin: *Theobroma cacao* is native to the Amazon basin; DNA evidence confirms an Amazonian origin with a strong pre-Columbian trade network (Mongabay 2024; archaeological cultivation records ~3,000+ years, with indigenous fermentation of a bitter ritual drink — the root of the word “chocolate”). Pará state carries the world's largest diversity of cacao germplasm (Portal Amazônia) — a product of várzea soils, climate and centuries of local selection.

Traditional landraces: the “common” cacao of Pará corresponds to the *forastero*/Amelonado-type landraces and named local varieties such as **Pará** and **Parazinho** (small fruit, strong regional adaptation, characteristic flavour used in artisanal chocolate). The farmers' “cacau comum” — seed-planted, seasonal, with a fruitless period — matches the traditional *forastero*/Pará-type habit.

Pará today: Brazil's leading cacao-producing state (with Rondônia and Bahia; Embrapa mapped ~121,000 ha across Pará and Rondônia; Pará + Bahia ≈ 96% of national output). Much new Pará planting is in agroforestry systems (SAFs) on former pasture — a model that fits the DAO's agroforestry/reforestation mission.

8 · Chemical differences (research findings)

Direct comparisons (Nacional vs CCN-51, fermented beans — Los Ríos, Ecuador, 2025): moisture, fat, caffeine and theobromine showed **no significant differences between varieties alone**, but the **genotype × location interaction was highly significant for total polyphenols (p = 0.003) and theobromine (p = 0.009)** — i.e. chemistry is driven by terroir interacting with genetics. Nacional from Montalvo peaked at 130.51 mg GAE/g polyphenols and up to 1.92% theobromine — the best functional profile of the study. Fat content did not differ significantly.

Brazilian clonal-cultivar study (MDPI IJMS 26(7):3386, 2025): CCN-51 scored 14.9% for cocoa flavour and 12.4% for bitterness — among the highest bitterness in the panel, with the **lowest sweetness scores** — versus BN 34. Polyphenols correlate positively with bitterness/adstringency and negatively with fruity notes (Luna et al. 2002). In a Pará pulp study (CCN-51 + CEPEC 2005 blend), pulp was low-calorie (56 kcal), 20.4 °Brix soluble solids, potassium-rich (128 mg/100 g).

Component (fermented bean)	Traditional/Nacional-type	CCN-51
Total polyphenols	Higher potential — up to 130.5 mg GAE/g (Montalvo); genotype×terroir dependent	Lower in most studies; bitterness/adstringency high
Theobromine	Up to 1.92% (Nacional Montalvo)	No significant difference vs Nacional alone
Fat (cocoa butter)	~50% typical; no significant difference	No significant difference (stable)
Caffeine	No significant difference	No significant difference
Sensory correlation	Fruity/floral potential, lower bitterness	Bitterness 12.4%, cocoa flavour 14.9%, low sweetness

9 · Taste profiles

CCN-51: historically maligned — “more bitter and less nuanced”, the poster child of bulk cacao. Early reviews were poor, but **skilled fermentation can unlock red-berry, tropical-fruit and soft-caramel notes** (Coeur de Xocolat). Cup character: pronounced cocoa flavour, high bitterness, low sweetness; moderate acidity; adstringency notable at high polyphenol levels. In blends it adds body and yield-economics.

Traditional Pará cacao: the Amazonian landraces (Parazinho, Pará, várzea natives) are the basis of Brazil's fine-flavour craft-chocolate boom — sensory work by UFPA maps notes of **fruity, floral, herbal, woody, nutty and caramel** with balanced acidity — what “cacao fino” certification looks for (low bitterness/adstringency, distinct pleasant flavour). Terroir (floodplain sediment, climate) drives the aromatic differentiation the state markets internationally.

10 · Sources

- Coeur de Xocolat — “CCN 51 Explained: Origins, Flavour, and Farmer Impact” (history, witches' broom, flavour with good fermentation).
- To'ak Chocolate — “The Different Grades of Ecuadorian Cacao” (genetic composition 1.1% Nacional / 45.4% IMC / 22.2% Criollo / 21.5% Amelonado / 3.9% Contamana / 2.5% Purús / 2.1% Maraño; 70% of Ecuador production; 90% of plantings).
- The C-Spot — CCN-51 strain atlas (IMC 67 / ICS 95 / O-1 pedigree; 1965 Finca Sofia; unpatented after Castro's death).
- MDPI IJMS 26(7):3386 (2025) — “Elite Cacao Clonal Cultivars... Rebuild the Cocoa Industry in Brazil” (organoleptic: cocoa flavour 14.9%, bitterness 12.4%, low sweetness).
- PPGPV-UESC (B. H. C. Porto) — clonal cultivar productivity in southern Bahia (CCN-51 2,143 → 2,955 kg/ha years 6–7; RNC/MAPA clonal list).
- Revista Alfa (2025) — “Caracterización química del cacao Nacional y CCN51 en Los Ríos, Ecuador” (polyphenols 130.51 mg GAE/g Nacional Montalvo; theobromine 1.92%; genotype×location $p=0.003/0.009$).
- UFPA / UFBA — sensory profiling of Pará nibs (fruity, floral, nutty notes; fine-cacao criteria) and cacao pulp composition (CCN-51+CEPEC 2005: 56 kcal, 20.4 °Brix, K 128 mg/100 g).
- Mongabay Brasil (2024) — DNA confirms Amazonian origin of cacao + pre-Columbian trade network; cultivation history in Brazilian Amazon since 17th c. colonization records.
- Portal Amazônia — “Pará tem a maior diversidade e variedade de sementes de cacau no mundo”; terroir of várzea; native cacao description.
- Embrapa — mapping of ~121,000 ha of cacao in Pará + Rondônia; geography of Amazon cacao production.
- ResearchGate — “Caracterização de variedades tradicionais de cacau (Theobroma cacao L.) — sul da Bahia” (varieties Pará and Parazinho).
- Field evidence: video clips IMG_7654/7669/7672/7673 (transcription + Grok vision phenotype analysis, Aug 2026).